$$p^*_{\text{MLE}} = \bar{x} = 0$$

In other words, it estimates the coin is most likely to have no chance of head. This is absurd.

- In practice, the Bayesian statistical framework (https://en.wikipedia.org/wiki/Bayesian_inference) is one way to address this kind of issue.

5 Linear Regression Part 2

5.1 Statistical Assumption

- Until now, we have framed the linear regression model as a mathematical optimization problem to find the optimal parameters $\beta_0^*, \dots, \beta_d^*$ which minimize the mean squared error MSE

$$\min_{\beta_0,\dots,\beta_d} \underbrace{\frac{1}{n}\sum_{i=1}^{n} \big(y_i - (\beta_0 + \beta_1 x_{i1} + \cdots + \beta_d x_{id})\big)^2}_{\text{MSE}}$$

Now let us reframe linear regression as a parametric statistical model along with its statistical inference perspective i.e. point estimate, confidence interval, and hypothesis testing.

- Let $(\vec{x}_1, y_1), \dots, (\vec{x}_n, y_n)$ be n pairs of observed data points with d-dimensional explanatory variable $\vec{x}_i = (x_{i1}, \dots, x_{id}) \in \mathbb{R}^d$. The statistical model assumption of linear regression is

$$Y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_d x_{id} + \epsilon_i \quad \text{where} \quad \epsilon_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$$

Equivalently, we have

$$Y_i \overset{\text{indep}}{\sim} \mathcal{N}(\beta_0 + \beta_1 x_{i1} + \cdots + \beta_d x_{id}, \sigma^2) \quad \text{for each } i = 1, \dots, n$$

In matrix form, it is a n-dimensional multivariate normal distribution

$$\vec{Y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix} \sim \mathcal{N}_n(X\vec{\beta}, \sigma^2 I_n)$$

with mean vector and covariance matrix

$$\mathbb{E}(\vec{Y}) = X\vec{\beta} = \underbrace{\begin{pmatrix} 1 & x_{11} & \dots & x_{1d} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \dots & x_{nd} \end{pmatrix}}_{\text{design matrix}} \times \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_d \end{pmatrix} \in \mathbb{R}^n \qquad \text{Cov}(\vec{Y}) = \sigma^2 I_n = \underbrace{\begin{pmatrix} \sigma^2 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma^2 \end{pmatrix}}_{\text{diagonal matrix}} \in \mathbb{R}^{n \times n}$$

This says:

1. **Deterministic explanatory variable:** Explanatory variables $\vec{x}_i$ is fixed / deterministic / not a random variable in this statistical model.
2. **Random response variable:** Response variable Y_i is viewed as a random variable because the noise / error term ϵ_i is a random variable. Each observed response value y_i is a realization of its random variable Y_i.
3. **Normally distributed:** Since the amount of random error ϵ_i is normally distributed, so is Y_i. (Note this does not imply a histogram of the observed response values $y_1, \dots, y_n$ together would have a bell-shaped curve because each Y_i has a different expectation $\mathbb{E}(Y_i)$.)
4. **Linear:** There is a linear relationship between the expected response $\mathbb{E}(Y_i)$ and the explanatory variable $\vec{x}_i$.

$$\mathbb{E}(Y_i) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_d x_{id}$$

For each ith observed data point, since $\vec{x}_i$ can be different, so could $\mathbb{E}(Y_i)$. The commonalty across the n data points is the form of the linear equation along with the $\beta_0, \ldots, \beta_d$ parameters.

5. **Constant variance:** The random errors ϵ_i's have constant variance across all observed data points.

$$\text{Var}(Y_i) = \text{Var}(\epsilon_i) = \sigma^2 \quad \text{for each } i = 1, \ldots, n$$

6. **Independence:** Since the random errors ϵ_i's are independent random variables, the response random variables $Y_1, \ldots, Y_n$ are also independent too.

- By reframing as a statistical model, what do we gain beyond our previous focus on just MSE optimization?
  - **Model selection:** It provides a principled way to perform explanatory variable / model selection based on statistical hypothesis testing e.g. $H_0 : \beta_j = 0$ for usefulness of x_j.
  - **Confidence interval for prediction:** For an unseen value of explanatory variable $\vec{x}$, besides a point estimate $\hat{y}$ of predicted value, we can construct $(1-\alpha)\%$ confidence interval in the form $\hat{y} \pm t_{1-\alpha/2}\,(\text{s.e.})$.

5.2 Chi-squared $\mathcal{X}^2_k$

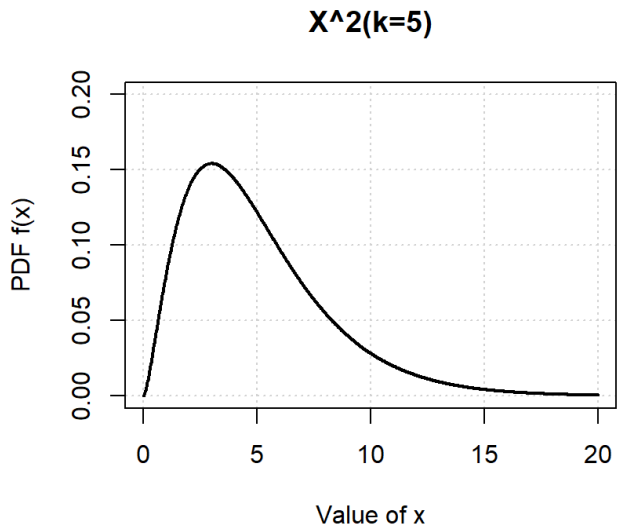
- A chi-squared random variable $X \sim \mathcal{X}^2_k$ with **degrees of freedom** parameter $k \in \mathbb{N}$ is a sum of k squared independent standard normal random variables i.e.

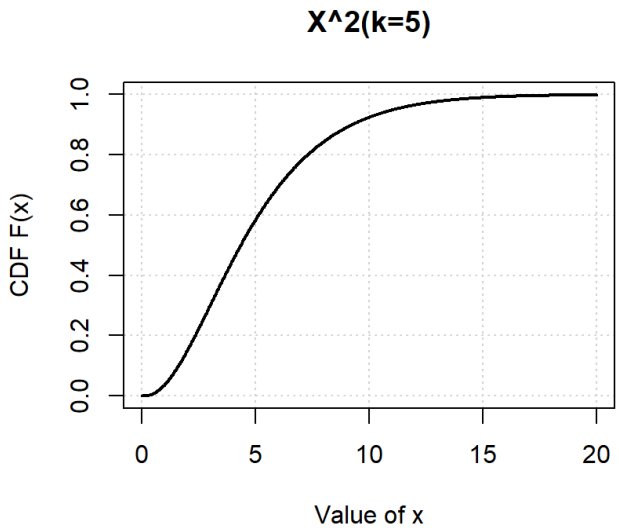
$$X = Z_1^2 + \cdots + Z_k^2$$

  with $Z_1, \ldots, Z_k \sim \mathcal{N}(0, 1)$.
- It is always non-negative $X \geq 0$ i.e. has support on the non-negative real number line $[0, \infty)$.
- Probability density function f_X is

$$f_X(x) = \frac{x^{k/2-1} e^{-x/2}}{2^{k/2}\Gamma(k/2)} \quad \text{for } x > 0$$

  where $\Gamma(k/2)$ is the complicated gamma function (https://en.wikipedia.org/wiki/Gamma_function). Its cumulative distribution function F_X is an integral but does not have a simple closed-form formula.






```
# Simulate X^2(5) observations
k <- 5
rchisq(5, df = k)
```

```
[1] 3.391515 3.884121 6.998860 5.034811 6.481215
```

- Probability density function f_X is right-skewed and unimodal with mode at $x = k - 2$ when $k \geq 2$.
- By linearity of expectation and variance,

$$\mathbb{E}(X) = \mathbb{E}(Z_1^2) + \cdots + \mathbb{E}(Z_k^2) = k$$
$$\mathrm{Var}(X) = \mathrm{Var}(Z_1^2) + \cdots + \mathrm{Var}(Z_k^2) = 2k$$

- In practice, chi-squared distribution is mostly used in statistical hypothesis testing e.g. **chi-squared test** of independence for contingency table, goodness-of-fit test of observed data against some probability distribution, or likelihood ratio test LRT to select between two nested statistical models later.

5.3 Estimation of $\vec{\beta}$ and σ^2

- Based on the statistical assumption of linear regression model, the likelihood function is

$$\begin{aligned} L(\beta_0, \ldots, \beta_d, \sigma^2) &= P\left(Y_1 = y_1, \ldots, Y_n = y_n\right) = \prod_{i=1}^{n} P(Y_i = y_i) \\ &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y_i - (\beta_0 + \beta_1 x_{i1} + \cdots + \beta_d x_{id}))^2} \\ &= \left(2\pi\sigma^2\right)^{-n/2} e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - (\beta_0 + \beta_1 x_{i1} + \cdots + \beta_d x_{id}))^2} \end{aligned}$$

  The log-likelihood function is

$$\begin{aligned} l(\beta_0, \ldots, \beta_d, \sigma^2) &= \log L(\beta_0, \ldots, \beta_d, \sigma^2) \\ &= -\frac{n}{2}\log\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - (\beta_0 + \beta_1 x_{i1} + \cdots + \beta_d x_{id}))^2 \end{aligned}$$

- By the likelihood principle, to maximize the likelihood function L with respect to $\vec{\beta}$, it suffices to maximize the 2nd term above, which is equivalent to minimizing the MSE. Therefore, the maximum likelihood estimate MLE for $\vec{\beta}$ is

$$\vec{\beta}^*_{\text{MLE}} = \vec{\beta}^*_{\text{LS}} = (X^T X)^{-1} X^T \vec{y}$$

  is exactly the same as the optimal parameters $\vec{\beta}^*_{\text{LS}}$ via least squares previously.
- Later, we will perform statistical inference on the model parameter $\vec{\beta}$. To do so, we need to know the reference probability distribution of the estimator $\vec{\beta}^*_{\text{MLE}}$. Since the response vector $\vec{Y} \sim \mathcal{N}_n(X\vec{\beta}, \sigma^2 I_n)$ is multivariate normally distributed and $\vec{\beta}^*_{\text{MLE}} = (X^T X)^{-1} X^T \vec{Y}$ is a matrix multiplication (a kind of affine transformation) of $\vec{Y}$, it is also multivariate normally distributed. Now the mean vector and covariance matrix of $\vec{\beta}^*_{\text{MLE}}$ are

$$\mathbb{E}(\vec{\beta}^*_{\text{MLE}}) = (X^T X)^{-1} X^T \, \mathbb{E}(\vec{Y}) = (X^T X)^{-1} X^T X \vec{\beta} = \vec{\beta} \in \mathbb{R}^d$$

$$\mathrm{Var}(\vec{\beta}^*_{\text{MLE}}) = (X^T X)^{-1} X^T \, \mathrm{Var}(\vec{Y}) \, X (X^T X)^{-T} = \sigma^2 (X^T X)^{-1} \in \mathbb{R}^{d \times d}$$

  Therefore,

$$\vec{\beta}^*_{\text{MLE}} \sim \mathcal{N}_d(\vec{\beta},\, \sigma^2 (X^T X)^{-1})$$

  In particular, $\vec{\beta}^*_{\text{MLE}}$ is an unbiased estimator for $\vec{\beta}$.

- The maximum likelihood estimate MLE for σ^2 is simply the (unadjusted) sample variance of the residuals, which is same as the mean squared error MSE i.e.

$$\hat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^{n}\underbrace{\left(y_i - \hat{y}_i\right)}_{\text{error } e_i}^2 = \frac{SS_{\text{error}}}{n} = \text{MSE}$$

- However, $\hat{\sigma}^2_{\text{MLE}}$ is not an unbiased estimator for σ^2. To see this, let us consider the simplest linear regression model with only the intercept term i.e.

$$Y_i = \beta_0 + \epsilon_i \quad \text{where} \quad \epsilon_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$$

  It is equivalent to the statistical model

$$Y_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\beta_0, \sigma^2)$$

  which estimates the mean and variance parameters of a normal distribution based on observed data. We know an unbiased estimator for σ^2 requires the $n-1$ adjustment i.e.

$$\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y}_n)^2$$

- Back to multiple linear regression, let $p = 1 + d = \dim(\vec{\beta})$ be the total number of intercept and slope parameters in the model. This is called the **model degrees of freedom**. (Note p does not count the σ^2 parameter of the error term.) Now using the fact that

$$\frac{SS_{\text{error}}}{\sigma^2} = \sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{\sigma}\right)^2 \sim \mathcal{X}^2_{n-p}$$

  has a chi-squared distribution with $n-p$ degrees of freedom, we have $\mathbb{E}(SS_{\text{error}}/\sigma^2) = n - p$. Rewriting it yields

$$\mathbb{E}\left(\frac{SS_{\text{error}}}{n-p}\right) = \sigma^2$$

  This motivates us to consider the following (adjusted) unbiased estimator for σ^2:

$$\hat{\sigma}^2 = \frac{SS_{\text{error}}}{n-p} = \frac{1}{n-p}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

  The quantity $n-p$ is called the **residual degrees of freedom**. (This generalizes the above (adjusted) sample variance estimator beyond $p = 1$.)

```
x <- c(0.4, 0.8, 1.2, 1.7, 2.1)
y <- c(2, 5, 4.3, 4.2, 8)
model <- lm(y ~ x)

# Compute unscaled covariance matrix (X^T X)^{-1}
X <- model.matrix(y ~ x)
solve(t(X) %*% X)
summary(model)$cov.unscaled
```

```
            (Intercept)          x
(Intercept)   1.0302376 -0.6695464
x            -0.6695464  0.5399568
            (Intercept)          x
(Intercept)   1.0302376 -0.6695464
x            -0.6695464  0.5399568
```

```
# Compute model degrees of freedom p
model$qr$rank
summary(model)$df[1]
```

```
[1] 2
[1] 2
```

```
# Compute residual degrees of freedom n - p
df.residual(model)
model$df.residual
summary(model)$df[2]
length(y) - model$qr$rank
```

```
[1] 3
[1] 3
[1] 3
[1] 3
```

```
# Compute adjusted unbiased estimated sigma
summary(model) # Residual standard error
summary(model)$sigma
sqrt(deviance(model) / df.residual(model)) # sqrt(SSE / (n - p))
```

```


Call:
lm(formula = y ~ x)

Residuals:
      1       2       3       4       5
-0.5410  1.4309 -0.2972 -1.6823  1.0896

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)    1.513      1.488   1.017   0.3841
x              2.570      1.077   2.386   0.0971 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.466 on 3 degrees of freedom
Multiple R-squared:  0.6549,    Adjusted R-squared:  0.5399
F-statistic: 5.694 on 1 and 3 DF,  p-value: 0.09707

[1] 1.465819
[1] 1.465819
```

5.4 Hypothesis Testing

- In statistical hypothesis testing, we have two competing hypotheses: the **null hypothesis** H_0 and the **alternative hypothesis** H_1 about the unknown true parameter(s) of our statistical model. Based on some observed data points, we would like to decide if the evidence supports H_0 or H_1.
- *Example:* To determine if a coin is fair or not, we toss the coin n times and observe $x_1, \ldots, x_n \in \{\text{head}, \text{tail}\}$. Our two hypotheses could be

$$H_0 : \text{Coin is fair i.e. } P(\text{head}) = 0.5$$
$$H_1 : \text{Coin is unfair i.e. } P(\text{head}) \neq 0.5$$

  In this case, H_0 is a **simple / point hypothesis** since it consists of only one possible value $p = 0.5$ of the model parameter. On the other hand, H_1 is a **composite hypothesis** which consists of many possible values of the model parameter i.e. all $p \neq 0.5$.
- If our observed evidence supports H_0 and it turns out H_0 is actually true, then we are correct. Similarly for H_1. Otherwise, we make a mistake.
  - **Type I error**: Reject H_0 (or accept H_1) when H_0 is true
  - **Type II error**: Retain / fail to reject H_0 when H_1 is true
- In practice, the null hypothesis H_0 typically represents an existing view / conventional belief / baseline / no effect e.g. a new medicine is not useful. The alternative hypothesis H_1 typically represents a challenge perspective / new theory / impactful treatment e.g. a new medicine improves health. In this case, a type I error is usually considered worse than a type II error.
- To conduct a hypothesis test:
  1. Summarize the observed evidence into one number called a **test statistic**. In general, a **statistic** is a quantity derived from the observed data.
  2. This test statistic is viewed as an observed value of a random variable. It has a reference probability distribution.
  3. Compute **p-value** of this test statistic using its reference probability distribution under the assumption that the null hypothesis H_0 is true.

$$\text{p-value} = P_{H_0}(\text{such or more extreme outcome than our observed evidence})$$

  4. This p-value represents the level of “surprise”. If p-value is small ≈ 0, then our observed test statistic evidence is very extreme so it is unlikely to occur under H_0. This suggests that H_0 might be false.
- Most common misinterpretation of p-value:

$$\text{p-value} \neq P(H_0 \text{ is true})$$

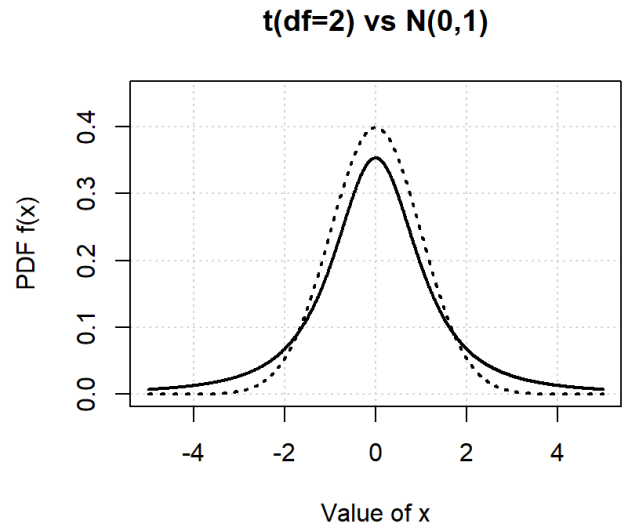
  The null hypothesis H_0 is either true or false. It is unknown but not a random event. So p-value is not the probability of such.

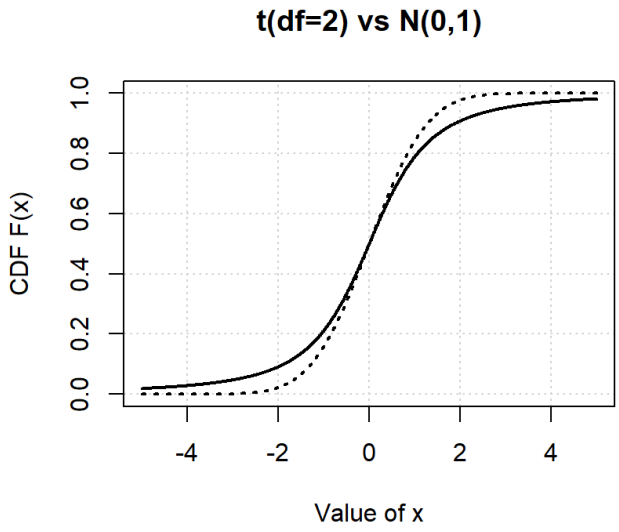
5.5 Student’s t t_ν

- A Student’s t random variable $X \sim t_\nu$ with **normality parameter** or **degrees of freedom** parameter $\nu > 0$ has probability density function

$$f_X(x) = \frac{1}{\sqrt{\nu} B(1/2, \nu/2)} \left(1 + \frac{t^2}{\nu}\right)^{-(\nu+1)/2}$$

  Here $B(1/2, \nu/2)$ is the complicated Beta function (https://en.wikipedia.org/wiki/Beta_function) constant just to make the area under the probability density function f_X curve equals to 1. Its cumulative distribution function $F_X(x)$ is an integral but does not have a simple closed-form formula.






```
# Simulate t(2) observations
nu <- 2
rt(5, df = nu)
```

```
[1]  0.4601457 -0.9656381  0.6731015  0.6987576 -0.1030360
```

- Probability density function f_X is also a unimodal **bell-shaped** curve. Since it is symmetric around $x = 0$, the mean / median / mode are all equal.

$$\mathbb{E}(X) = 0 \qquad \mathrm{Var}(X) = \begin{cases} \frac{\nu}{\nu-2} & \text{if } \nu > 2 \\ \infty & \text{otherwise} \end{cases}$$

- It is a **fat-tailed** distribution (relative to the thin tail of the normal distribution) with kurtosis

$$\mathrm{Kurt}(X) = \begin{cases} 3 + \frac{6}{\nu-4} & \text{if } \nu > 4 \\ \infty & \text{otherwise} \end{cases}$$

  which is greater than $\mathrm{Kurt}(\mathcal{N}(\mu, \sigma^2)) = 3$. Smaller ν value yields heavier tail.
- In general, a probability distribution with high kurtosis and fat tail means it is not a **rare event** to observe sample data with large (positive or negative) magnitude.
- For $\nu \leq 2$, large positive and negative data are frequent enough that the Student's t distribution has infinite variance $\mathrm{Var}(X) = \infty$.
- In practice, the Student's t distribution is often used in robust regression when outliers are more frequent. For example, in financial markets, large profits and losses are more frequent than would be described by a normal distribution.
- Student's t distribution plays an important role in one-sample and two-sample mean comparison hypothesis testing and confidence interval construction later.
- As $\nu \to \infty$, the Student's t distribution converges to the standard normal distribution

$$t_\nu \overset{d}{\approx} \mathcal{N}(0, 1) \quad \text{for large } \nu$$

5.6 F-distribution F_{d_1, d_2}

- An F-distributed random variable $X \sim F_{d_1, d_2}$ with degree of freedom parameters $d_1, d_2 \in \mathbb{N}$ is a ratio of two independent $\mathcal{X}^2$ chi-squared random variables i.e.

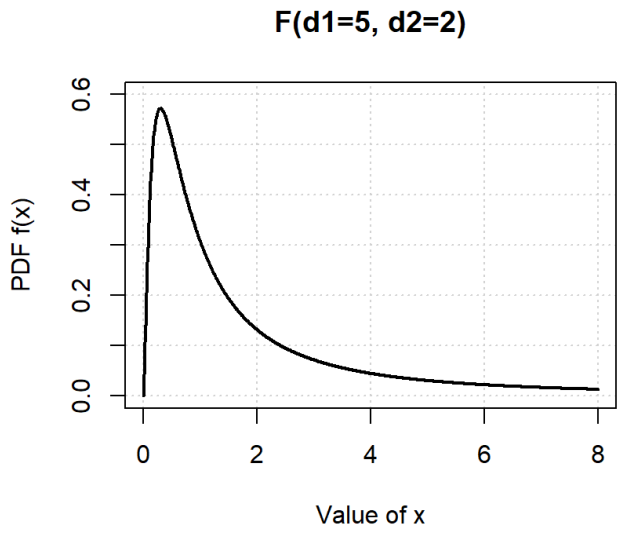
$$X = \frac{C_1/d_1}{C_2/d_2}$$

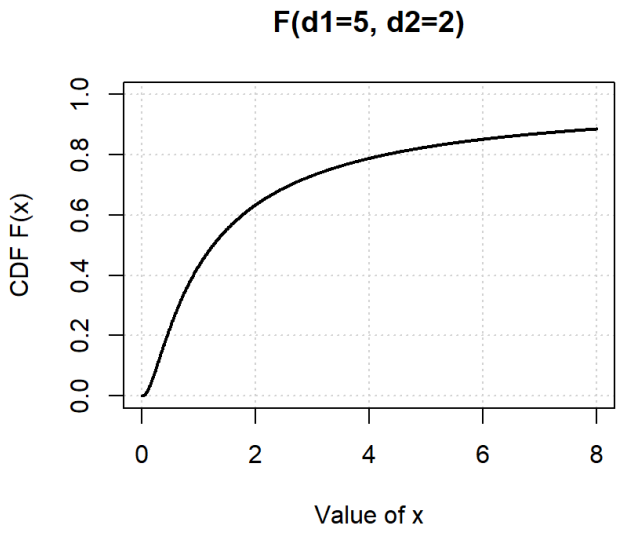
with $C_1 \sim \mathcal{X}^2_{d_1}$ and $C_2 \sim \mathcal{X}^2_{d_2}$.

- It is always positive $X > 0$ i.e. has support on the positive real number line $(0, \infty)$.
- Probability density function f_X is

$$f_X(x) = \frac{1}{\mathrm{B}\left(\frac{d_1}{2}, \frac{d_2}{2}\right)} \left(\frac{d_1}{d_2}\right)^{d_1/2} x^{d_1/2-1} \left(1 + \frac{d_1}{d_2} x\right)^{-(d_1+d_2)/2} \quad \text{for } x > 0$$

where $\mathrm{B}\left(\frac{d_1}{2}, \frac{d_2}{2}\right)$ is the complicated beta function (https://en.wikipedia.org/wiki/Beta_function). Its cumulative distribution function F_X is an integral but does not have a simple closed-form formula.






```
# Simulate F(5,2) observations
d1 <- 5
d2 <- 2
rf(5, df1 = d1, df2 = d2)
```

```
[1] 17.377144  0.882152  1.536231  4.487365  1.777715
```

- Probability density function f_X is right-skewed and unimodal.
- Mean $\mathbb{E}(X)$ is

$$\mathbb{E}(X) = \frac{d_2}{d_2 - 2} \quad \text{for } d_2 > 2$$

- As $d_2 \to \infty$, the denominator of an F-distributed random variable becomes

$$C_2/d_2 = \frac{1}{d_2} \sum_{i=1}^{d_2} Z_i^2 \to \mathbb{E}(Z_i^2) = 1 \quad \text{as } d_2 \to \infty$$

based on the law of large numbers LLN (https://en.wikipedia.org/wiki/Law_of_large_numbers). In this case, the F-distributed random variable is

$$X = \frac{C_1/d_1}{C_2/d_2} \approx \frac{C_1}{d_1} \overset{d}{=} \frac{1}{d_1} \mathcal{X}^2_{d_1}$$

i.e. it converges to a scaled $\mathcal{X}^2$-distribution with d_1 degrees of freedom (with scaling of $1/d_1$). In other words,

$$F_{d_1,d_2} \overset{d}{\approx} \frac{1}{d_1}\mathcal{X}^2_{d_1} \quad \text{for large } d_2$$

- In practice, F-distribution is mostly used in statistical hypothesis testing e.g. **F-test** to select between two nested statistical models later.

5.7 Model Selection with F-Test and t-Test

- In machine learning modeling, there is always a trade-off between goodness-of-fit of a model to the training data vs **model complexity**. A complex model (e.g. too many explanatory variables, categorical variable with too many levels i.e. high cardinality, or highly nonlinear) might overfit / memorize the observed data and fail to recognize the general pattern for unseen data. **Model selection** strives to find the simplest model that fits the observed data adequately.
- In statistical linear regression modeling, one main tool for model selection is to conduct hypothesis tests called the **F-test**. It is specifically for comparing a larger model vs a smaller model that is nested within. (Technically, a **nested model** is obtained by imposing some linear constraint(s) to restrict the parameters $\beta_0, \ldots, \beta_d$ of the larger model.) Here are some examples of nested models:
    1. Decide if one (or more) explanatory variable(s) should be included in a model or not e.g.

    $$\begin{gathered} Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon \quad \text{vs} \quad Y = \beta_0 + \beta_1 x_1 + \epsilon \\ H_0 : \beta_2 = 0 \quad \text{vs} \quad H_1 : \beta_2 \neq 0 \end{gathered}$$

    2. Decide if a higher degree term is needed in a polynomial regression e.g.

    $$\begin{gathered} Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \epsilon \quad \text{vs} \quad Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \epsilon \\ H_0 : \beta_3 = 0 \quad \text{vs} \quad H_1 : \beta_3 \neq 0 \end{gathered}$$

    3. Decide if an interaction term is useful in a model or not e.g.

    $$\begin{gathered} Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \epsilon \quad \text{vs} \quad Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon \\ H_0 : \beta_3 = 0 \quad \text{vs} \quad H_1 : \beta_3 \neq 0 \end{gathered}$$

    4. Decide if two (or more) levels of a categorical variable should be combined or not e.g.

    $$\begin{gathered} Y = \beta_0 + \beta_1 x_1 + \beta_2 \mathbf{1}_{\{\text{Windy}\}} + \beta_3 \mathbf{1}_{\{\text{Cloudy}\}} + \epsilon \quad \text{vs} \quad Y = \beta_0 + \beta_1 x_1 + \beta_2 \mathbf{1}_{\{\text{Windy}\}} + \beta_2 \mathbf{1}_{\{\text{Cloudy}\}} + \epsilon \\ H_0 : \beta_2 = \beta_3 \quad \text{vs} \quad H_1 : \beta_2 \neq \beta_3 \end{gathered}$$

    5. **Omnibus F-test:** Decide if any explanatory variable is useful in the model at all e.g.

    $$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \epsilon \quad \text{vs} \quad Y = \beta_0 + \epsilon$$

    $$H_0 : \beta_1 = \beta_2 = \beta_3 = 0 \quad \text{vs} \quad H_1 : \text{ some } \beta_j \neq 0$$

- Let $SS_{\text{error}}^{\text{large}}$ and $SS_{\text{error}}^{\text{small}}$ be the residual sums of squares (represent goodness-of-fit) of the larger and smaller nested model repsectively. Let

$$p^{\text{large}} > p^{\text{small}}$$

be their model degrees of freedom (represent model complexity). Since the larger model is more flexible and can behave like the smaller nested model if it wants to, it would not fit the observed data worse than the smaller model. Therefore, it must have higher R^2 and lower SS_{error} and MSE i.e.

$$SS_{\text{error}}^{\text{large}} < SS_{\text{error}}^{\text{small}}$$

- We are interested in whether the reduction in SS_{error} is significant enough to worth the increase in the number of model parameters p. To do so, let us consider the F-test statistic

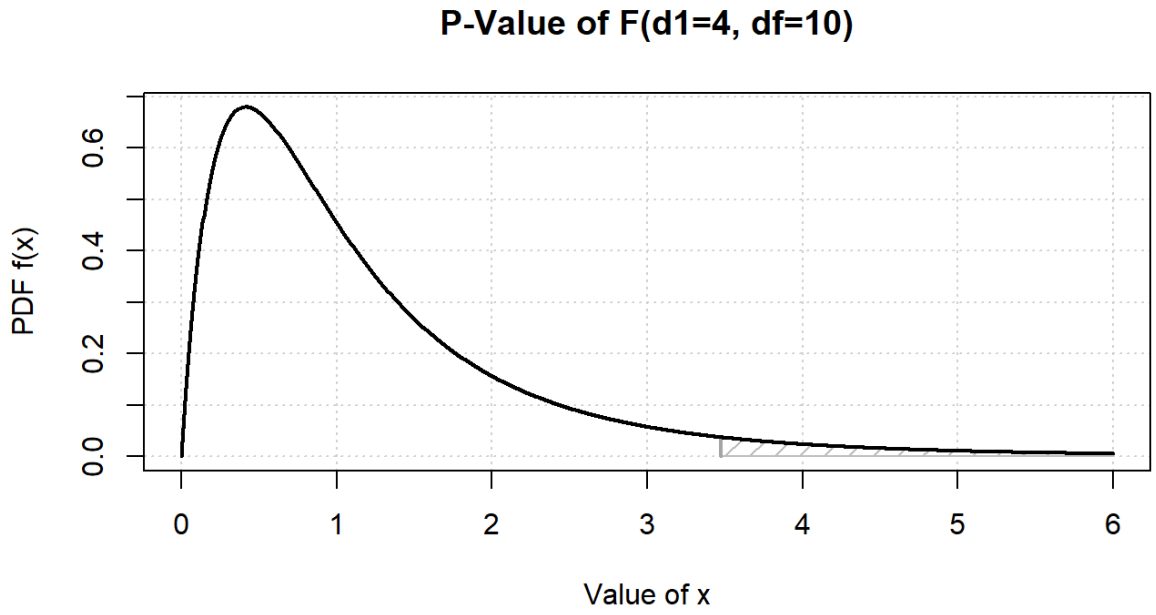
$$\mathrm{f} = \left(\frac{SS_{\text{error}}^{\text{small}} - SS_{\text{error}}^{\text{large}}}{(n - p^{\text{small}}) - (n - p^{\text{large}})} \right) \Big/ \left(\frac{SS_{\text{error}}^{\text{large}}}{n - p^{\text{large}}} \right) = \left(\frac{SS_{\text{error}}^{\text{small}} - SS_{\text{error}}^{\text{large}}}{p^{\text{large}} - p^{\text{small}}} \right) \Big/ \left(\frac{SS_{\text{error}}^{\text{large}}}{n - p^{\text{large}}} \right)$$

It represents the relative reduction in SS_{error} benchmarked against the number of model parameters.

- Under the null hypothesis H_0 that the smaller nested model suffices, we have the reference probability distribution

$$\mathrm{f} \sim F_{(p^{\text{large}} - p^{\text{small}}),\, (n - p^{\text{large}})} \quad \text{under } H_0$$

If H_0 is false, then the smaller model is insufficient so it yields large $SS_{\text{error}}^{\text{small}} \gg SS_{\text{error}}^{\text{large}}$. As a result, the F-test statistic $\mathrm{f} \gg 0$. Therefore, we compute p-value based on the right-tail of the corresponding F-distribution and reject H_0 if the value of f is large. If we want significance level at $\alpha\%$, then we use the $(1-\alpha)\%$-tile $\mathrm{f}_{1-\alpha}$ as the rejection threshold.



- *Example:* Below is an example to perform the omnibus F-test found in `summary(model)` for the null hypothesis $H_0 : \beta_1 = \ldots = \beta_d = 0$.

```
# Omnibus F-test
set.seed(1)

n <- 100
x1 <- runif(n)
x2 <- runif(n)
y <- 1 + 0 * x1 + 2 * x2 + rnorm(n, sd = 0.5)

large.model <- lm(y ~ x1 + x2)
summary(large.model)
```

```
Call:
lm(formula = y ~ x1 + x2)

Residuals:
    Min      1Q  Median      3Q     Max
-0.9290 -0.3084 -0.0716  0.2676  1.1659

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.99532    0.13953   7.133 1.77e-10 ***
x1          -0.03463    0.18172  -0.191    0.849
x2           2.00718    0.17889  11.220  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4838 on 97 degrees of freedom
Multiple R-squared:  0.5648,    Adjusted R-squared:  0.5558
F-statistic: 62.94 on 2 and 97 DF,  p-value: < 2.2e-16
```

```
small.model <- lm(y ~ 1) # H0: b1 = b2 = 0
summary(small.model)
```

```
Call:
lm(formula = y ~ 1)

Residuals:
     Min       1Q   Median       3Q      Max
-1.44921 -0.57786 -0.02299  0.56204  1.50374

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.01604    0.07259   27.77   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7259 on 99 degrees of freedom
```

```
anova(small.model, large.model) # Not to be confused with aov()
```

```
Analysis of Variance Table

Model 1: y ~ 1
Model 2: y ~ x1 + x2
  Res.Df   RSS Df Sum of Sq      F    Pr(>F)
1     99 52.16
2     97 22.70  2     29.46 62.943 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Compute F-statistic and p-value manually
(ss.error.large <- deviance(large.model))
p.large <- 3

(ss.error.small <- deviance(small.model))
p.small <- 1

(f <- ((ss.error.small - ss.error.large) / (p.large - p.small)) / (ss.error.large / (n - p.large)))

(p.value <- 1 - pf(f, p.large - p.small, n - p.large))
```

```
[1] 22.70011
[1] 52.16008
[1] 62.94282
[1] 0
```

- *Example:* Below is an example to perform F-test for the null hypothesis $H_0 : \beta_j = 0$ to decide to select the explanatory variable x_j or not.

```
set.seed(1)

n <- 100
x1 <- runif(n)
x2 <- runif(n)
y <- 1 + 0 * x1 + 2 * x2 + rnorm(n, sd = 0.5)

large.model <- lm(y ~ x1 + x2)
summary(large.model)
```

```

Call:
lm(formula = y ~ x1 + x2)

Residuals:
    Min      1Q  Median      3Q     Max 
-0.9290 -0.3084 -0.0716  0.2676  1.1659 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  0.99532    0.13953   7.133 1.77e-10 ***
x1          -0.03463    0.18172  -0.191    0.849    
x2           2.00718    0.17889  11.220  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4838 on 97 degrees of freedom
Multiple R-squared:  0.5648,    Adjusted R-squared:  0.5558 
F-statistic: 62.94 on 2 and 97 DF,  p-value: < 2.2e-16
```

```
small.model <- update(large.model, . ~ . - x1) # Modify the model formula and refit
summary(small.model)
```

```
Call:
lm(formula = y ~ x2)

Residuals:
     Min       1Q   Median       3Q      Max
-0.93937 -0.30670 -0.06959  0.26911  1.17215

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.9777     0.1039   9.408 2.34e-15 ***
x2            2.0066     0.1780  11.274  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4814 on 98 degrees of freedom
Multiple R-squared:  0.5646,    Adjusted R-squared:  0.5602
F-statistic: 127.1 on 1 and 98 DF,  p-value: < 2.2e-16
```

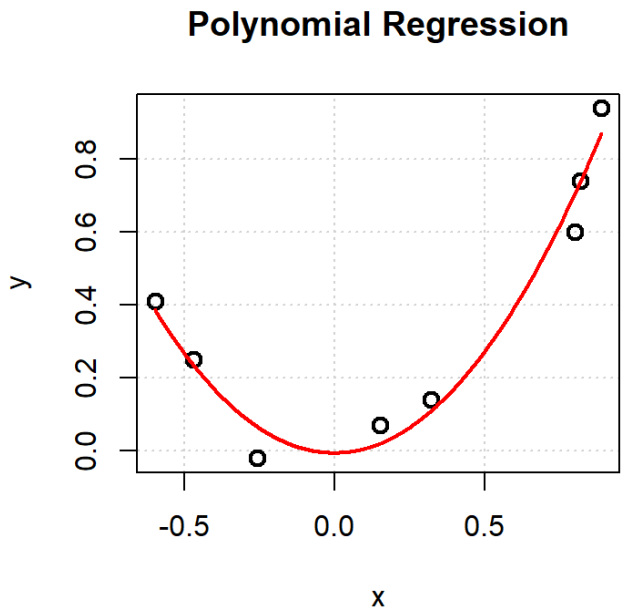
```
anova(small.model, large.model)
```

```
Analysis of Variance Table

Model 1: y ~ x2
Model 2: y ~ x1 + x2
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1     98 22.709
2     97 22.700  1 0.0084993 0.0363 0.8493
```

- *Example:* Below is an example to perform sequential F-tests to decide the degree of a polynomial regression. This works because `anova(model)` will perform an incremental F-test for each additional explanatory variable $x_j = x^j$ that appears in the model formula.

```
x <- c(-0.47, -0.26, 0.15, 0.82, -0.6, 0.80, 0.89, 0.32)
y <- c(0.25, -0.02, 0.07, 0.74, 0.41, 0.60, 0.94, 0.14)
```




```
model <- lm(y ~ x + I(x ^ 2) + I(x ^ 3) + I(x ^ 4))
summary(model)
```

```
Call:
lm(formula = y ~ x + I(x^2) + I(x^3) + I(x^4))

Residuals:
        1         2         3         4         5         6         7         8
 0.064391 -0.047974  0.001283  0.031879 -0.028174 -0.055377  0.014006  0.019966

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.03396    0.05198   0.653    0.560
x            0.19594    0.15809   1.239    0.303
I(x^2)       0.32877    0.50045   0.657    0.558
I(x^3)      -0.85442    0.49158  -1.738    0.181
I(x^4)       1.68878    0.90312   1.870    0.158

Residual standard error: 0.06304 on 3 degrees of freedom
Multiple R-squared:  0.9855,    Adjusted R-squared:  0.9662
F-statistic: 50.97 on 4 and 3 DF,  p-value: 0.004327
```

```
anova(model)
```

```
Analysis of Variance Table

Response: y
          Df  Sum Sq Mean Sq  F value   Pr(>F)
x          1 0.37449 0.37449  94.2444 0.002321 **
I(x^2)     1 0.42035 0.42035 105.7851 0.001960 **
I(x^3)     1 0.00143 0.00143   0.3593 0.591136
I(x^4)     1 0.01389 0.01389   3.4967 0.158262
Residuals  3 0.01192 0.00397
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- If we use `poly()` for feature engineering, then R views all polynomial degree terms together as one group and considers the null hypothesis $H_0 : \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0$ instead. It does not perform individual sequential F-test.

```
anova(lm(y ~ poly(x, 5, raw = TRUE)))
```

```
Analysis of Variance Table

Response: y
                      Df  Sum Sq Mean Sq F value  Pr(>F)
poly(x, 5, raw = TRUE)  5 0.81701 0.16340   64.33 0.01538 *
Residuals               2 0.00508 0.00254
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- *Example:* Below is an example to perform F-test to decide to select (all levels of) a categorical variable or not.

```
set.seed(2)

n <- 100
temperature <- runif(n, min = 0, max = 100)
weather <- sample(c('Sunny', 'Windy', 'Cloudy'), n, replace = TRUE)
weather.offset <- numeric(length(temperature))
weather.offset[weather == 'Sunny'] <- 1
weather.offset[weather == 'Windy'] <- -2
weather.offset[weather == 'Cloudy'] <- -2
feel <- temperature + weather.offset + rnorm(n, sd = 5)

large.model <- lm(feel ~ temperature + weather)
summary(large.model)
```

```
Call:
lm(formula = feel ~ temperature + weather)

Residuals:
    Min      1Q  Median      3Q     Max
-9.7972 -4.0050  0.7037  3.4849 13.2348

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  -3.34919    1.23868  -2.704  0.00811 **
temperature   1.01901    0.01717  59.348  < 2e-16 ***
weatherSunny  3.62232    1.21055   2.992  0.00352 **
weatherWindy  0.79247    1.29197   0.613  0.54108
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.018 on 96 degrees of freedom
Multiple R-squared:  0.9747,    Adjusted R-squared:  0.9739
F-statistic:  1232 on 3 and 96 DF,  p-value: < 2.2e-16
```

- Since the categorical variable `weather` has 3 levels, the larger model is

$$Y = \beta_0 + \beta_1 \, (\text{temperature}) + \beta_2 \, \mathbf{1}_{\{\text{Sunny}\}} + \beta_3 \, \mathbf{1}_{\{\text{Windy}\}} + \epsilon$$

  In this case, the null hypothesis is

$$H_0 : \beta_2 = \beta_3 = 0$$

  i.e. none of `Sunny` or `Windy` is different from the baseline categorical level of `Cloudy` in the intercept term.

```
small.model <- lm(feel ~ temperature)
summary(small.model)

anova(small.model, large.model)
```

```
Call:
lm(formula = feel ~ temperature)

Residuals:
     Min       1Q   Median       3Q      Max
-10.5479  -3.8432   0.6347   3.0454  14.8833

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.03001    1.00953  -2.011   0.0471 *
temperature  1.02575    0.01758  58.360   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.221 on 98 degrees of freedom
Multiple R-squared:  0.972, Adjusted R-squared:  0.9717
F-statistic:  3406 on 1 and 98 DF,  p-value: < 2.2e-16

Analysis of Variance Table

Model 1: feel ~ temperature
Model 2: feel ~ temperature + weather
  Res.Df    RSS Df Sum of Sq      F   Pr(>F)
1     98 2671.1
2     96 2416.9  2    254.16 5.0477 0.008234 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- *Example:* Below is an example to perform F-test to decide to combine some levels of a categorical variable or not. Continuing from the larger model

$$Y = \beta_0 + \beta_1 \left(\text{temperature}\right) + \beta_2 \, \mathbf{1}_{\{\text{Sunny}\}} + \beta_3 \, \mathbf{1}_{\{\text{Windy}\}} + \epsilon$$

  Now the null hypothesis is $H_0 : \beta_3 = 0$ i.e. `Windy` should not have its own $\beta_3 \, \mathbf{1}_{\{\text{Windy}\}}$ term and should be collapsed into the intercept term shared by `Cloudy` instead.

```
weather.c <- weather
weather.c[weather.c == 'Cloudy'] = 'Cloudy or Windy'
weather.c[weather.c == 'Windy'] = 'Cloudy or Windy'
small.model <- lm(feel ~ temperature + weather.c)
summary(small.model)
```

```
Call:
lm(formula = feel ~ temperature + weather.c)

Residuals:
    Min      1Q  Median      3Q     Max
-9.8551 -3.7343  0.6198  3.4539 13.2811

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)    -2.90973    1.00720  -2.889  0.00477 **
temperature     1.01792    0.01702  59.797  < 2e-16 ***
weather.cSunny  3.24233    1.03668   3.128  0.00233 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.001 on 97 degrees of freedom
Multiple R-squared:  0.9746,    Adjusted R-squared:  0.9741
F-statistic:  1860 on 2 and 97 DF,  p-value: < 2.2e-16
```

```
anova(small.model, large.model)
```

```
Analysis of Variance Table

Model 1: feel ~ temperature + weather.c
Model 2: feel ~ temperature + weather
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1     97 2426.4
2     96 2416.9  1    9.4722 0.3762 0.5411
```

- To decide to select an explanatory variable x_j or not, we focus on the null hypothesis $H_0 : \beta_j = 0$. Previously, we conduct this hypothesis testing using the F-test, which evaluates if the reduction in SS_{error} by the larger model with $\beta_j \neq 0$ is worth the increase in the $+1$ number of model parameters vs the smaller model with $\beta_j = 0$.
- Besides the F-test, an alternative (and mathematically equivalent) way to conduct this hypothesis testing is to use the t-test to evaluate directly if the model parameter estimate $\hat{\beta}_j$ is statistically significantly different from 0 or not. To do so:
    1. Compute standard error of the $\hat{\beta}_j$ estimator

    $$\text{s.e.}(\hat{\beta}_j) = \hat{\sigma}\sqrt{(X^T X)^{-1}_{ii}}$$

    This comes from the multivariate normal distribution of the MLE estimator $\vec{\beta}^*_{\text{MLE}}$.
    2. Compute t-statistic

    $$t_j = \frac{\hat{\beta}_j - 0}{\text{s.e.}(\hat{\beta}_j)} = \frac{\hat{\beta}_j}{\hat{\sigma}\sqrt{(X^T X)^{-1}_{ii}}} \sim t_{n-p} \quad \text{under } H_0 : \beta_j = 0$$

    3. Compute (two-sided) p-value using the reference probability distribution t_{n-p} with $\nu = n - p$ residual degrees of freedom. This kind of hypothesis testing is called a **Wald test**.
- The `Pr(>|t|)` column of `summary(model)` provides a quick overview of individual hypothesis tests $H_0 : \beta_j = 0$ for each model parameter β_j.

```
set.seed(1)

n <- 100
x1 <- runif(n)
x2 <- runif(n)
y <- 1 + 0 * x1 + 2 * x2 + rnorm(n, sd = 0.5)

model <- lm(y ~ x1 + x2)
summary(model)
```

```

Call:
lm(formula = y ~ x1 + x2)

Residuals:
    Min      1Q  Median      3Q     Max 
-0.9290 -0.3084 -0.0716  0.2676  1.1659 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  0.99532    0.13953   7.133 1.77e-10 ***
x1          -0.03463    0.18172  -0.191    0.849    
x2           2.00718    0.17889  11.220  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4838 on 97 degrees of freedom
Multiple R-squared:  0.5648,    Adjusted R-squared:  0.5558 
F-statistic: 62.94 on 2 and 97 DF,  p-value: < 2.2e-16
```

```
# Step 1: Compute standard error for beta
X <- model.matrix(y ~ x1 + x2)
C.unscaled <- solve(t(X) %*% X) # Also summary(model)$cov.unscaled

sigma <- summary(model)$sigma
C <- sigma ^ 2 * C.unscaled # Cov(estimated beta)  Also vcov(model)
(s.e.beta <- sqrt(diag(C))) # s.e.(beta)
```

```
(Intercept)          x1          x2 
  0.1395314   0.1817239   0.1788948 
```

```
# Step 2: Compute t-statistic
beta <- coef(model)
(t.stat <- beta / s.e.beta)
```

```
(Intercept)          x1          x2 
  7.1332728  -0.1905738  11.2198767 
```

```
# Step 3: Compute (two-sided) p-value using reference t-distribution
(p.value <- 2 * pt(abs(t.stat), df = df.residual(model), lower.tail = FALSE))
```

```
 (Intercept)           x1           x2
1.771804e-10 8.492578e-01 3.196083e-19
```

- Why is this t-test equivalent to the corresponding F-test? First, the F-test statistic and the t-statistic are related by

$$\mathrm{f} = t_j^2$$

```
large.model <- lm(y ~ x1 + x2)
small.model <- lm(y ~ x2) # H0: b1 = 0
(f.test <- anova(small.model, large.model))

(f.stat <- f.test$F[2]) # F-test statistic
```

```
Analysis of Variance Table

Model 1: y ~ x2
Model 2: y ~ x1 + x2
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1     98 22.709
2     97 22.700  1 0.0084993 0.0363 0.8493
[1] 0.03631838
```

```
summary(large.model)

t.stat <- coef(summary(large.model))[2,3] # t-statistic
t.stat ^ 2
```

```

Call:
lm(formula = y ~ x1 + x2)

Residuals:
    Min      1Q  Median      3Q     Max
-0.9290 -0.3084 -0.0716  0.2676  1.1659

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.99532    0.13953   7.133 1.77e-10 ***
x1          -0.03463    0.18172  -0.191    0.849
x2           2.00718    0.17889  11.220  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4838 on 97 degrees of freedom
Multiple R-squared:  0.5648,    Adjusted R-squared:  0.5558
F-statistic: 62.94 on 2 and 97 DF,  p-value: < 2.2e-16

[1] 0.03631838
```

- Secondly, the p-value computed from the right-tail of the $F_{1,n-p}$ distribution is same as the two-sided p-value computed from the t_{n-p} distribution.

```
pf(f.stat, df1 = 1, df2 = df.residual(large.model), lower.tail = FALSE)
2 * pt(abs(t.stat), df = df.residual(large.model), lower.tail = FALSE)
```

```
[1] 0.8492578
[1] 0.8492578
```

- The F-test is more general. Its null hypothesis can involve more complex relationship across multiple model parameters β_j's. On the other hand, a t-test only concerns about if a particular model parameter $b_j = 0$ or not.
- For each t-test, the interpretation of the null hypothesis $H_0 : \beta_j = 0$ is "if all other explanatory variables are already present in the model, then is β_j still useful or not?" This means:
    1. The set of individual t-tests are separate and not related to each other e.g. not sequential.
    2. If an explanatory variable is added to or removed from the model, then the t-tests and p-values for other explanatory variables usually change.

```
x1 <- c(-0.55, 0.86, 1.31, 1.48, -0.10, 2.41, 1.33, 0.04, -1.59, 1.19)
x2 <- c(0.22, 0.36, 1.52, 1.73, 0.80, 1.82, 1.73, 0.74, -1.36, 0.68)
y <- c(0.44, 3.48, 5.23, 8.20, 4.02, 13.99, 8.64, 4.06, -6.36, 2.66)

summary(lm(y ~ x1 + x2))
```

```

Call:
lm(formula = y ~ x1 + x2)

Residuals:
    Min      1Q  Median      3Q     Max 
-2.6879 -0.5385  0.1398  0.7764  3.1883 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)  
(Intercept)   0.5827     0.8455   0.689   0.5129  
x1            1.7067     1.1484   1.486   0.1808  
x2            3.3549     1.3880   2.417   0.0463 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.881 on 7 degrees of freedom
Multiple R-squared:  0.905, Adjusted R-squared:  0.8778 
F-statistic: 33.33 on 2 and 7 DF,  p-value: 0.0002645
```

```
summary(lm(y ~ x1))
```

```
Call:
lm(formula = y ~ x1)

Residuals:
    Min      1Q  Median      3Q     Max
-4.0746 -1.7899  0.6044  1.9162  2.6572

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.7792     0.8685   2.049 0.074670 .
x1            4.1642     0.6765   6.155 0.000272 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.383 on 8 degrees of freedom
Multiple R-squared:  0.8257,    Adjusted R-squared:  0.8039
F-statistic: 37.89 on 1 and 8 DF,  p-value: 0.0002724
```

```
summary(lm(y ~ x2))
```

```
Call:
lm(formula = y ~ x2)

Residuals:
    Min      1Q  Median      3Q     Max
-2.8120 -0.9142 -0.3908  0.4043  4.3937

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.1668     0.8560   0.195     0.85
x2            5.1810     0.6924   7.483 7.04e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.018 on 8 degrees of freedom
Multiple R-squared:  0.875, Adjusted R-squared:  0.8594
F-statistic:    56 on 1 and 8 DF,  p-value: 7.04e-05
```

- If there is (near) multicollinearity among some explanatory variables, then the t-test's p-values might seem misleading (unless you understand its proper interpretation).

```
set.seed(1)

n <- 100
x1 <- runif(n)
x2 <- x1 + rnorm(n, sd = 0.001)
y <- 1 + 1 * x1 + 1 * x2 + rnorm(n, sd = 0.1)

cor(x1, x2)
```

```
[1] 0.9999939
```

```
summary(lm(y ~ x1 + x2)) # Both x1 and x2 have large p-values
```

```

Call:
lm(formula = y ~ x1 + x2)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.28311 -0.07273 -0.00537  0.06338  0.23359 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  1.01305    0.02319  43.688   <2e-16 ***
x1          -6.11730   11.34084  -0.539    0.591    
x2           8.09674   11.33723   0.714    0.477    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1056 on 97 degrees of freedom
Multiple R-squared:  0.9626,    Adjusted R-squared:  0.9618 
F-statistic:  1248 on 2 and 97 DF,  p-value: < 2.2e-16
```

```
summary(lm(y ~ x1))
```

```

Call:
lm(formula = y ~ x1)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.288231 -0.068074 -0.007586  0.061531  0.241729 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  1.01160    0.02304   43.90   <2e-16 ***
x1           1.98197    0.03957   50.09   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1054 on 98 degrees of freedom
Multiple R-squared:  0.9624,    Adjusted R-squared:  0.962 
F-statistic:  2509 on 1 and 98 DF,  p-value: < 2.2e-16
```

```
summary(lm(y ~ x2))
```

```
Call:
lm(formula = y ~ x2)

Residuals:
      Min        1Q    Median        3Q       Max
-0.286993 -0.068484 -0.007379  0.061259  0.237965

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.01191    0.02301   43.98   <2e-16 ***
x2           1.98142    0.03951   50.15   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1052 on 98 degrees of freedom
Multiple R-squared:  0.9625,    Adjusted R-squared:  0.9621
F-statistic:  2515 on 1 and 98 DF,  p-value: < 2.2e-16
```

- The t-test for an individual categorical level is a comparison with the baseline categorical level only (in case when the model has an intercept term). If you would like to decide to select the entire categorical variable or not, then do a F-test using the `anova()` function instead.

```
set.seed(2)

n <- 100
temperature <- runif(n, min = 0, max = 100)
weather <- sample(c('Sunny', 'Windy', 'Cloudy'), n, replace = TRUE)
weather.offset <- numeric(length(temperature))
weather.offset[weather == 'Sunny'] <- 1
weather.offset[weather == 'Windy'] <- -2
weather.offset[weather == 'Cloudy'] <- -2
feel <- temperature + weather.offset + rnorm(n, sd = 5)

summary(lm(feel ~ temperature + weather))
```

```
Call:
lm(formula = feel ~ temperature + weather)

Residuals:
    Min      1Q  Median      3Q     Max
-9.7972 -4.0050  0.7037  3.4849 13.2348

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  -3.34919    1.23868  -2.704  0.00811 **
temperature   1.01901    0.01717  59.348  < 2e-16 ***
weatherSunny  3.62232    1.21055   2.992  0.00352 **
weatherWindy  0.79247    1.29197   0.613  0.54108
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.018 on 96 degrees of freedom
Multiple R-squared:  0.9747,    Adjusted R-squared:  0.9739
F-statistic:  1232 on 3 and 96 DF,  p-value: < 2.2e-16
```

5.8 Information Criteria and Adjusted R^2

- Since the F-test (and hence t-test) for model selection is only applicable to comparison between two nested models, we need some other metrics to select between two competing models in general:
    1. **Akaike Information Criterion AIC:** It is motivated by information theory (https://en.wikipedia.org/wiki/Information_theory) to measure the distance between the fitted model vs the unknown true model. We want to minimize

    $$\text{AIC} = -2\ \log(L) + 2\,p$$

    Here L is the likelihood function value of the fitted linear regression model evaluated at the MLEs from the observed data (so the 1st term represents goodness-of-fit). Moreover, $p = d + 2$ is the total number of parameters in the model, which counts all of $\beta_0, \ldots, \beta_d, \sigma^2$ (so the 2nd term represents model complexity).
    2. **Bayes Information Criterion BIC:** It is defined based on the Bayesian framework (https://en.wikipedia.org/wiki/Bayesian_inference) to approximate the posterior probability (https://en.wikipedia.org/wiki/Posterior_probability) of the observed data under the fitted model. We want to minimize

    $$\text{BIC} = -2\ \log(L) + \log(n)\,p$$

    3. **Adjusted R^2:** Since the coefficient of determination R^2 never decreases with more explanatory variable(s) in a linear regression model, it cannot be used for model selection. To introduce a model complexity trade-off to the R^2 formula, we modify it by dividing each of its SS_{error} and SS_{total} by their corresponding residual degrees of freedom $n - p$ and $n - 1$ respectively. We want to maximize

    $$\text{adjusted } R^2 = 1 - \frac{SS_{\text{error}}/(n-p)}{SS_{\text{total}}/(n-1)} = 1 - \frac{\hat{\sigma}^2_{\text{model}}}{\hat{\sigma}^2_{\text{null}}} \in (-\infty, R^2]$$

    Unlike AIC and BIC defintions above, here $p = 1 + d$ does not count the σ^2 parameter for the variance of the noise term ϵ.

```
x <- c(-0.55, 0.86, 1.31, 1.48, -0.10, 2.41, 1.33, 0.04, -1.59, 1.19)
y <- c(0.44, 3.48, 5.23, 8.20, 4.02, 13.99, 8.64, 4.06, -6.36, 2.66)
model <- lm(y ~ x)

# Compute log-likelihood
logLik(model)
```

```
'log Lik.' -21.75862 (df=3)
```

```
# Compute log-likelihood manually
n <- length(y)
sigma.mle <- sqrt(mean(residuals(model) ^ 2)) # Unadjusted sigma from MLE
(log.L <- -(n / 2) * log(2 * pi * sigma.mle ^ 2) - 1 / (2 * sigma.mle ^ 2) * sum(residuals(model) ^ 2))
```

```
[1] -21.75862
```

```
# Compute AIC
AIC(model) # See also extractAIC()
```

```
[1] 49.51725
```

```
# Compute AIC manually
p <- 2 + 1 # Count all of b0, b1, and sigma
-2 * log.L + 2 * p
```

```
[1] 49.51725
```

```
# Compute BIC
AIC(model, k = log(n)) # -2 * log(L) + k * p where k = log(n)
```

```
[1] 50.425
```

```
# Compute BIC manually
-2 * log.L + log(n) * p
```

```
[1] 50.425
```

```
# Compute adjusted R^2
summary(model)
summary(model)$adj.r.squared

# Compute adjusted R^2 manually
ss.error <- sum(residuals(model) ^ 2)
ss.total <- sum((y - mean(y)) ^ 2)
p <- 2 # Count b0 and b1 but not sigma
(adj.r2 <- 1 - (ss.error / (n - p)) / (ss.total / (n - 1)))
```

```
Call:
lm(formula = y ~ x)

Residuals:
    Min      1Q  Median      3Q     Max
-4.0746 -1.7899  0.6044  1.9162  2.6572

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.7792     0.8685   2.049 0.074670 .
x             4.1642     0.6765   6.155 0.000272 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.383 on 8 degrees of freedom
Multiple R-squared:  0.8257,    Adjusted R-squared:  0.8039
F-statistic: 37.89 on 1 and 8 DF,  p-value: 0.0002724

[1] 0.8038707
[1] 0.8038707
```

- Now you should know the meaning and computation of every number shown in `summary(model)`.
- Which of these AIC, BIC, and adjusted R^2 metrics should we use in practice?
  - For AIC (and similarly BIC), different software tools might compute the log-likelihood values differently (up to an additive constant). Therefore, AIC values are intended to be compared relatively only (so the additive constant would be cancelled out anyway). An individual AIC value of a model might not have much meaning. In particular, there is no concept of a "large enough" AIC value that would indicate if a model is good enough or not.
  - For sufficiently large data set, we have $\log(n) > 2$. In this case, BIC penalizes model complexity more severely (i.e. prefers a simpler model) than AIC.
  - For adjusted R^2, it is essentially determined by the numerator $SS_{\text{error}}/(n-p) = \hat{\sigma}^2_{\text{model}}$ since the denominator $SS_{\text{total}}/(n-1)$ does not change across models.
  - For large data set with $n \gg 0$, $R^2 \approx$ adjusted R^2 since the ratio $(n-p)/(n-1) \approx 1$.
  - In theory, AIC and BIC have different asymptotic optimality behaviors. In practice, it is best to try all these metrics and see which models each metric selects, and also which explanatory variables are common or different among these selected models in order to obtain insight on the importance of each explanatory variable.

5.9 Forward, Backward, and Stepwise Regression

- If there are d possible explanatory variables $\vec{x} = (x_1, \ldots, x_d) \in \mathbb{R}^d$ (including variable transformation, polynomial term, interaction term, and one-hot encoding of categorical level), and each explanatory variable might or might not be selected, then there are 2^d possible linear regression models to be considered.
- It is usually too time-consuming and not practical to fit all of the models in order to perform F-test or compute AIC, BIC, and adjusted R^2 metrics to select the best model.
- We often use some **greedy algorithm (https://en.wikipedia.org/wiki/Greedy_algorithm)** to explore strategically only a subset of all possible models. (In general, a heuristic or procedure is "greedy" if it is myopic / short-sighted i.e. it makes incremental optimal choices locally with the hope that these individual choices together would be near the global optimum.) The most common algorithmic procedures for model selection are:
  1. **Forward selection**
     - Start with the simplest null linear regression model i.e. with intercept term only without any explanatory variable.
     - Iteratively, select the next most useful explanatory variable (that is not yet in the model) and include it in the model. Here "most useful" can be defined as the explanatory variable with the highest statistical

significance i.e. smallest p-value in a F-test or t-test against the current smaller nested model of selected explanatory variables, or would improve AIC / BIC / adjusted R^2 metric the most.
    - Stop when there is no more useful explanatory variable to be included into the model. This could be based on a significance level threshold on p-value e.g. $\alpha = 0.2$, or no more improvement in AIC / BIC / adjusted R^2 metric.
2. **Backward elimination**
    - Start with the most complex full linear regression model with all of the explanatory variables.
    - Iteratively, identify the least useful explanatory variable (that is still in the model) and remove it from the model.
    - Stop when there is no more useless explanatory variable to be removed from the model.
3. **Stepwise selection (https://en.wikipedia.org/wiki/Stepwise_regression)**
    - (This is just a hybrid of forward selection and backward elimination.)
    - Start with either the null or full linear regression model.
    - Iteratively, either select the most useful explanatory variable (that is not in the model, maybe removed previously) to be included in the model, or select the least useful explanatory variable (that is currently in the model) to be removed from the model.

- Which model selection procedure should we use?
    - Forward selection only selects explanatory variables one at a time. If there is a pair of explanatory variables x_1 and x_2 that are very useful together but each is individually not as useful as another explanatory variable x_3, then x_3 might be selected instead of x_1 and x_2. This is a typical shortcoming of greedy algorithm.
    - Backward elimination has a slight advantage that it can see how all the explanatory variables work together in the initial full model. So it can make better decision on which explanatory variable to be removed. However, if there are too many explanatory variables to begin with, then it might not even be feasible to fit the full model.
    - Both forward selection and backward elimination handle multicollinearity well. If an explanatory variable is redundant, then it is unlikely to be selected in forward selection, and it is likely to be removed in backward elimination.
    - Different model selection procedures might yield different final models. In practice, besides comparing the sets of explanatory variables in these final models, we also obtain insight about the relative importance of the explanatory variables from the order of selections and removals.
- In R, use `add1()`, `drop1()`, and `step()` functions for stepwise regression.

```
# Simulate some observed data
set.seed(1)

n <- 100
x1 <- runif(n)
x2 <- runif(n)
x3 <- x1 + x2 + rnorm(n, sd = 0.1) # Multicollinear with x1 and x2
y <- 1 + x1 + x2 + rnorm(n)

cor(x1 + x2, x3)
```

```
[1] 0.9702168
```

```
full.model <- lm(y ~ x1 + x2 + x3)
summary(full.model) # Large p-values due to multicollinearity
```

```
Call:
lm(formula = y ~ x1 + x2 + x3)

Residuals:
     Min       1Q   Median       3Q      Max 
-2.92961 -0.45913  0.00181  0.65546  2.61178 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   1.0381     0.3025   3.431 0.000887 ***
x1            1.4854     1.1619   1.278 0.204213    
x2            1.5706     1.1686   1.344 0.182105    
x3           -0.5380     1.1007  -0.489 0.626091    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.049 on 96 degrees of freedom
Multiple R-squared:  0.1222,    Adjusted R-squared:  0.09476 
F-statistic: 4.454 on 3 and 96 DF,  p-value: 0.00565
```

```
# One iteration of forward selection
null.model <- lm(y ~ 1)
add1(null.model, scope = ~ x1 + x2 + x3, k = 2, test = 'F') # Use k = log(n) for BIC
```

```
Single term additions

Model:
y ~ 1
       Df Sum of Sq    RSS    AIC F value    Pr(>F)    
<none>              120.32 20.495                      
x1      1    6.6544 113.66 16.805  5.7375 0.0185062 *  
x2      1    8.0292 112.29 15.588  7.0076 0.0094584 ** 
x3      1   12.6747 107.64 11.363 11.5394 0.0009858 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- Here `add1(..., test = 'F')` means R would report p-value based on F-test. However, the stepwise regression `step()` variable selection logic is always based on AIC (when `k = 2`) or BIC (when `k = log(n)`) information criteria metric instead.

```
# Forward selection
step(null.model, scope = ~ x1 + x2 + x3 , direction = 'forward', k = 2, trace = 1)

# One iteration of backward elimination
drop1(full.model, k = 2, test = 'F')

# Backward elimination
step(full.model, direction = 'backward', k = 2, test = 'F', trace = 1)
```

```
Start:  AIC=20.49
y ~ 1

       Df Sum of Sq    RSS    AIC
+ x3    1   12.6747 107.64 11.363
+ x2    1    8.0292 112.29 15.588
+ x1    1    6.6544 113.66 16.805
<none>              120.32 20.495

Step:  AIC=11.36
y ~ x3

       Df Sum of Sq    RSS    AIC
<none>              107.64 11.363
+ x2    1  0.229172 107.41 13.150
+ x1    1  0.039638 107.60 13.326

Call:
lm(formula = y ~ x3)

Coefficients:
(Intercept)           x3
     1.1315       0.9049

Single term deletions

Model:
y ~ x1 + x2 + x3
       Df Sum of Sq    RSS    AIC F value Pr(>F)
<none>              105.61 13.462
x1      1   1.79780 107.41 13.150  1.6341 0.2042
x2      1   1.98733 107.60 13.326  1.8064 0.1821
x3      1   0.26286 105.88 11.711  0.2389 0.6261
Start:  AIC=13.46
y ~ x1 + x2 + x3

       Df Sum of Sq    RSS    AIC F value Pr(>F)
- x3    1   0.26286 105.88 11.711  0.2389 0.6261
- x1    1   1.79780 107.41 13.150  1.6341 0.2042
- x2    1   1.98733 107.60 13.326  1.8064 0.1821
<none>              105.61 13.462

Step:  AIC=11.71
y ~ x1 + x2

       Df Sum of Sq    RSS    AIC F value   Pr(>F)
<none>              105.88 11.711
- x1    1    6.4096 112.29 15.588  5.8722 0.017236 *
- x2    1    7.7844 113.66 16.805  7.1317 0.008881 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Call:
lm(formula = y ~ x1 + x2)

Coefficients:
```

```
(Intercept)          x1          x2
      1.039       0.951       1.032
```

- Forward selection and backward elimination are conceptually simple, easy to implement, and generally applicable to any machine learning modeling given a model evaluation metric. However, in order to decide which explanatory variable should be added or removed, we need to fit up to d models. This might be too computationally-intensive. In modern machine learning, there are other more advanced techniques (e.g. LASSO (https://en.wikipedia.org/wiki/Lasso_(statistics))) that could perform similar kind of variable selection by fitting just one model.

5.10 Model Diagnostic

- A statistical procedure (such as confidence interval construction or hypothesis testing) is **properly calibrated** if its probabilistic claims are accurate e.g. a 95% confidence interval that indeed contains the true parameter 95% of the time, or a hypothesis test at $\alpha = 10\%$ significance level that indeed makes a type I mistake 10% of the time.
- Proper calibration requires a statistical model and its assumptions to be valid.
- Linear regression model diagnostics focus mostly on the residual values $r_1, \ldots, r_n$ of a fitted model.
- The following sections are in the order of practical importance from modern machine learning modeling perspective. Here are some considerations:
    - Predictive performance is usually the most important i.e. low MSE and high R^2 and adjusted R^2. Good predictive performance mostly comes from useful explanatory variables and feature engineering i.e. suitable transformation of explanatory variables $\vec{x}$ and response y.
    - For large data set, it is more likely to have outliers and anomalies. It also has higher statistical power to detect even small violation. Therefore, it is more difficult to satisfy all of the model assumptions.
    - Nonetheless, a fitted model that violates some model assumptions can still be useful e.g. to represent the relationship between the explanatory variables $\vec{x}$ and response y, or have good predictive performance.

5.10.1 Linear Relationship

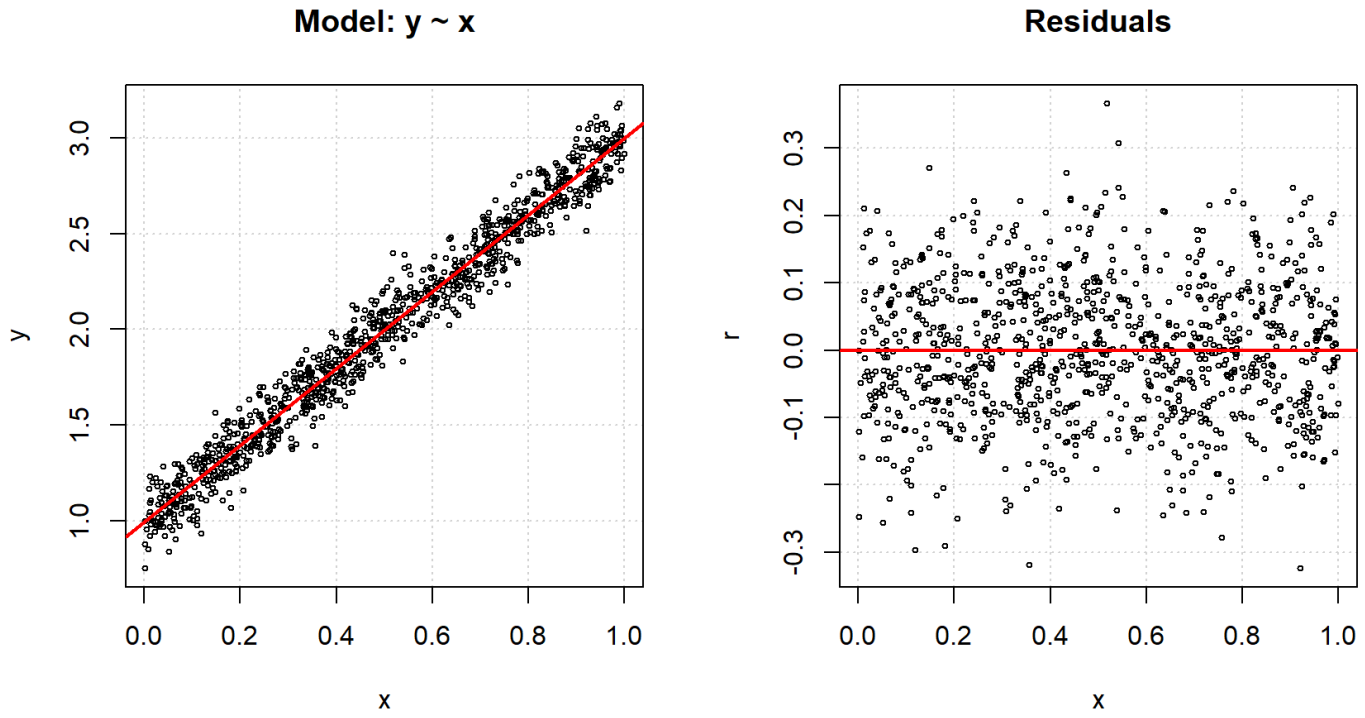
- The main assumption of linear regression model is that each explanatory variable x_j is linearly related to the response y. In this case, there should not be any systemic under- or over-prediction. The residuals should not have systemic positive and negative values in a plot vs each x_j.

```
n <- 1000
x <- runif(n)
y <- 1 + 2 * x + rnorm(n, sd = 0.1)

model <- lm(y ~ x)
r <- residuals(model)

plot(x, y, type = 'p', main = 'Linear Regression')
abline(model)

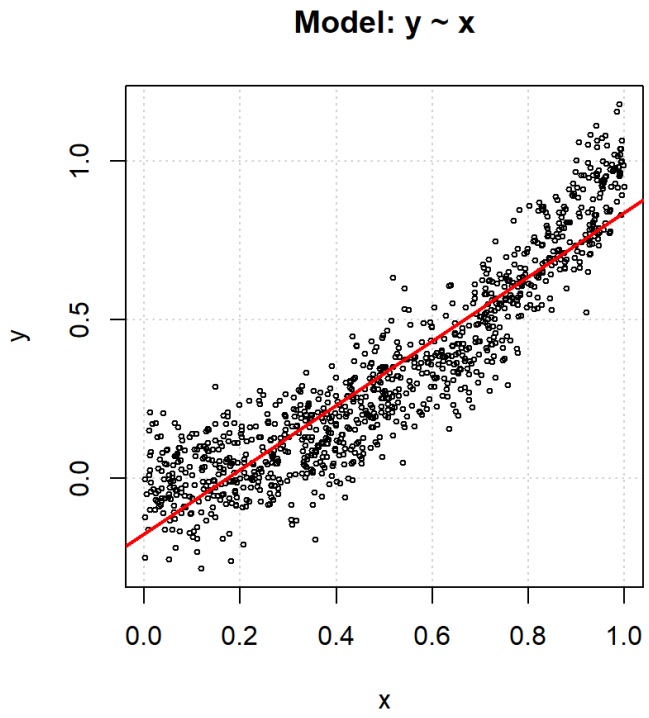
plot(x, r, type = 'p', main = 'Residuals')
abline(a = 0, b = 0)
```

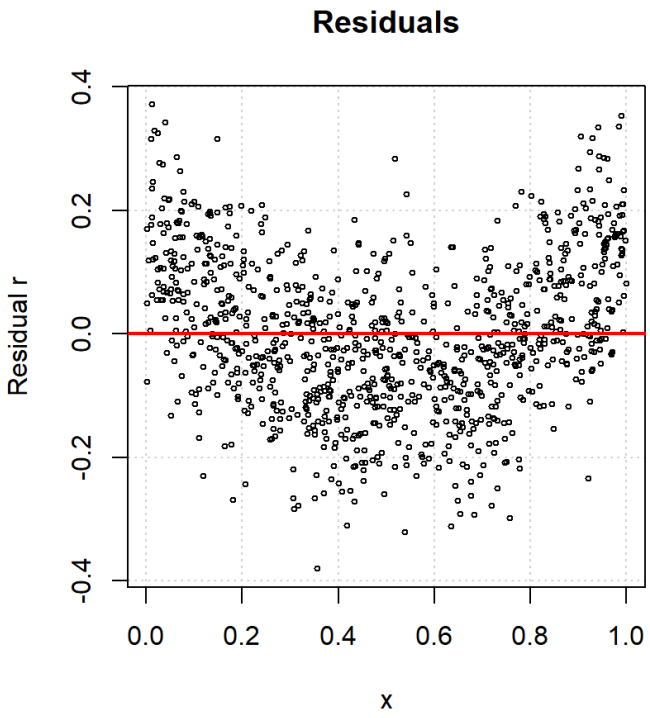


- If the linear assumption is violated, then the residuals might exhibit systemic positive and negative values in a plot vs some x_j.
- One remedy is to perform additional feature engineering e.g. introduce polynomial term (such as x^k), transformation (such as $\sqrt{x}$ or $\log x$), or interaction term to represent more complex mathematical relationship.

```
n <- 1000
x <- runif(n)
y <- x ^ 2 + rnorm(n, sd = 0.1)

model <- lm(y ~ x)
r <- residuals(model)
```





5.10.2 Homoscedasticity

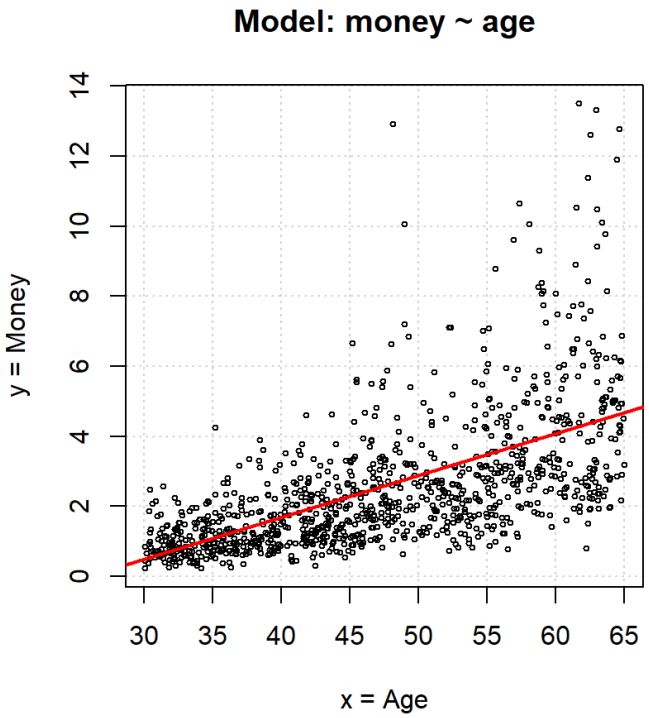
- **Homoscedasticity (https://en.wikipedia.org/wiki/Homoscedasticity_and_heteroscedasticity)** refers to the assumption of constant variance across all response values

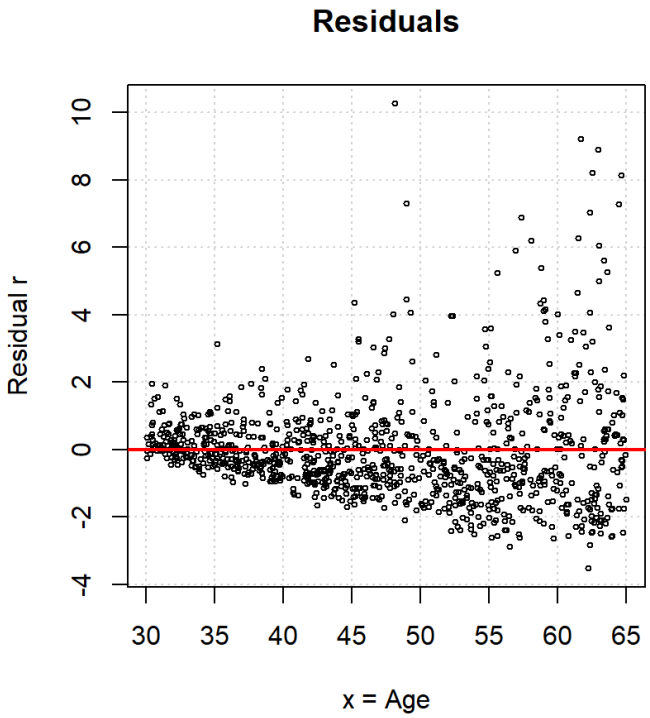
$$\mathrm{Var}(Y_1) = \ldots = \mathrm{Var}(Y_n) = \sigma^2$$

- Such assumption might be inappropriate due to the nature of the observed data. In many cases, the variance $\mathrm{Var}(Y_i)$ increases with the response value y_i. For example:
  - x = house size, y = house price
  - x = age of a person, y = net worth
  - x = distance of a taxi trip, y = trip duration
- If the constant variance assumption does not hold, then it is called **heteroscedasticity**.
- To identify heteroscedasticity, we usually plot the residual values of a fitted model against either the response variable y or any explanatory variable x_j and look for any systemic non-constant pattern.

```
n <- 1000
x <- runif(n, min = 30, max = 65)
b0 <- 0.2
b1 <- 1.05
y <- b0 * b1 ^ x * exp(rnorm(n, sd = 0.5))

model <- lm(y ~ x)
r <- residuals(model)
```





- There is also formal hypothesis test such as the Bresuch-Pagan test (https://en.wikipedia.org/wiki/Breusch%E2%80%93Pagan_test) in R's `lmtest::bptest()` for detecting heteroskedasticity.
- If heteroskedasticity is a result of the exponential growth (https://en.wikipedia.org/wiki/Exponential_growth) phenomenon with multiplicative error term ϵ (instead of additive error term i.e. $+\epsilon$)

  $$Y = \beta_0 \times (\beta_1)^x \times \epsilon$$

  then the variance

  $$\mathrm{Var}(Y) = \beta_0^2 \times (\beta_1)^{2x} \times \mathrm{Var}(\epsilon)$$

  indeed increases with y even if $\mathrm{Var}(\epsilon) = \sigma^2$ is constant. As a remedy, we take logarithm of both sides of the model equation so it becomes

  $$\log Y = \log(\beta_0) + \log(\beta_1)\, x + \log(\epsilon)$$

  Now it can be fitted as a linear regression model `lm(log(y) ~ x)`.
- Note that `lm(log(y) ~ x)` is not the same as `lm(y ~ exp(x))`. The latter has model equation

  $$Y = \beta_0 + \beta_1(e^x) + \epsilon$$

  which is the usual kind of linear regression model with an additive error term and an exponential variable transformation on x.
- Similarly, if the multiplicative relationship between the explanatory variable x and response variable y is

  $$Y = \beta_0 \times x^{\beta_1} \times \epsilon$$

  then taking logarithm would yield the linear regression model

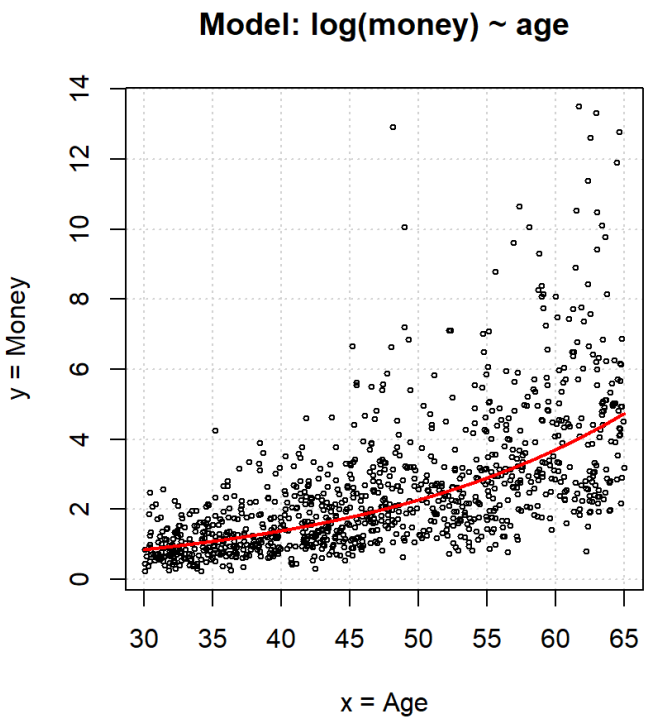
  $$\log Y = \log(\beta_0) + \beta_1 \log(x) + \log(\epsilon)$$

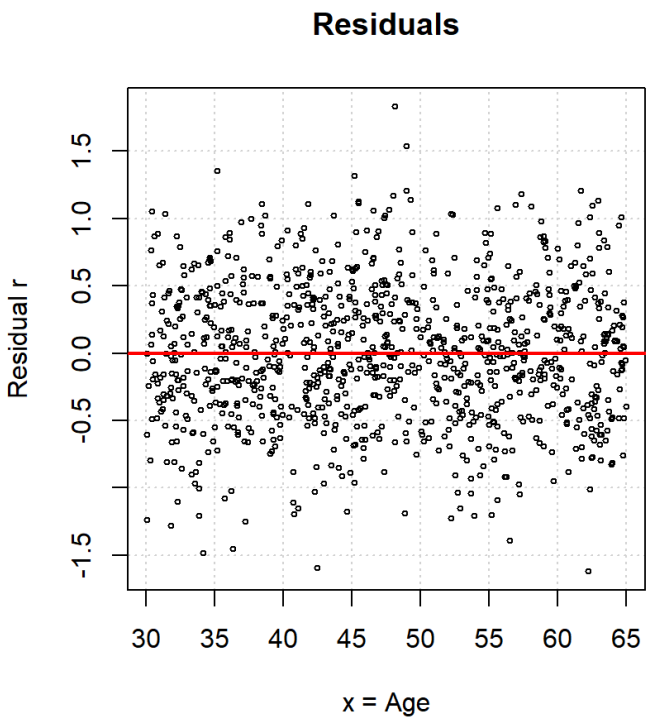
  i.e. `lm(log(y) ~ log(x))`.

```
model <- lm(log(y) ~ x)
r <- residuals(model)

exp(coef(model)) # Estimates of b0 and b1
```

```
(Intercept)           x
   0.194744    1.050318
```





- Not all heteroskedasticity could be addressed by transforming the response variable y. For example, $\log(y)$ or $\sqrt{y}$ is only applicable to $y > 0$.
- Another remedy is the **weighted least squares** (https://en.wikipedia.org/wiki/Weighted_least_squares) technique. If each error term ϵ_i has a different variance

$$\mathrm{Var}(\epsilon_i) = \sigma_i^2$$

but this variance is known (i.e. no longer an unknown model parameter), then the log-likelihood function of the statistical linear regression model is

$$l(\beta_0, \ldots, \beta_d) = (\text{constant}) - \underbrace{\frac{1}{2\sigma_i^2}}_{w_i} \sum_{i=1}^{n} (y_i - (\beta_0 + \beta_1 x_{i1} + \cdots + \beta_d x_{id}))^2$$

This is a weighted sum of squares of the residuals with weight $w_i = 1/(2\sigma_i^2)$. In general, the weighted-version of the mean squared error MSE is

$$\min_{\beta_0, \ldots, \beta_d} \underbrace{\frac{1}{n} \sum_{i=1}^{n} w_i \big(y_i - (\beta_0 + \beta_1 x_{i1} + \cdots + \beta_d x_{id})\big)^2}_{\text{weighted MSE}}$$

There exists closed-form formula and efficient computational technique to solve it for $\vec{\beta}^*$. Intuitively, if a data point has

high variance i.e. large σ_i^2, then it would have small weight w_i so the weighted least squares optimization focuses less on this data point.

5.10.3 Normality

- The "square" in mean squared error MSE and least squares LS are motivated by the normality assumption on the error term

$$\epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

and the corresponding mathematical form of the likelihood function.

- We assess the normality assumption of the (unobserved) ϵ_i's by examining the fitted residuals r_i's. (Strictly speaking, the r_i's are just a proxy. They are slightly correlated i.e. not independent. Their variances are also not exactly σ^2.)

$$\text{Var}(r_i) = \sigma^2(1 - h_{ii}) \neq \sigma^2 \quad \text{where} \quad H = X(X^TX)^{-1}X^T$$

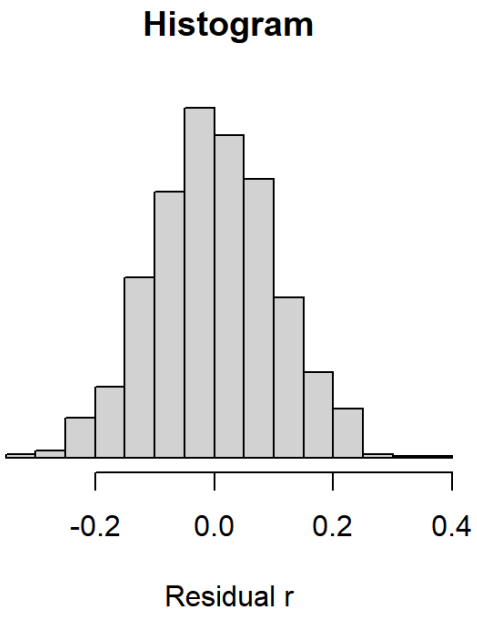
- Besides plotting a histogram of the residuals r_i's and look for a bell-shape, we can also use the **Q-Q plot** (https://en.wikipedia.org/wiki/Q%E2%80%93Q_plot). It provides a clearer visual comparison between the observed data sample (in our case, the residuals) vs the quantiles of a given reference probability distribution.
- In R, use `qqnorm()` (or more generally `qqplot()`) and `qqline()`.

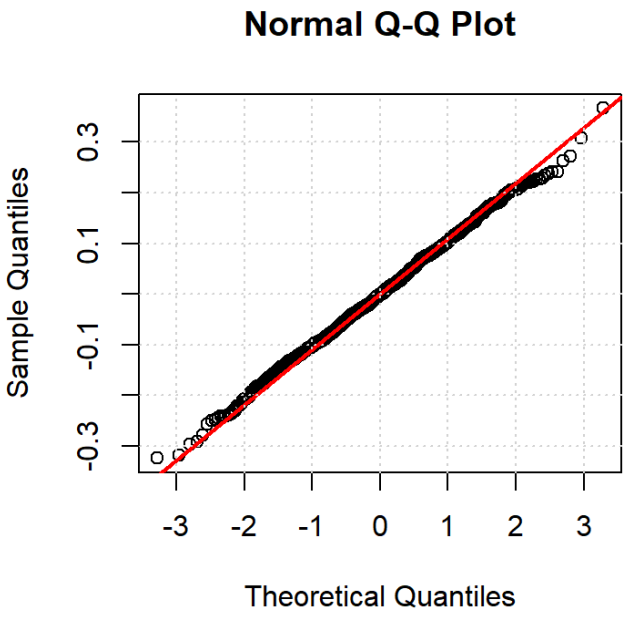
```
n <- 1000
x <- runif(n)
y <- 1 + 2 * x + rnorm(n, sd = 0.1)

model <- lm(y ~ x)
r <- residuals(model)

hist(r)

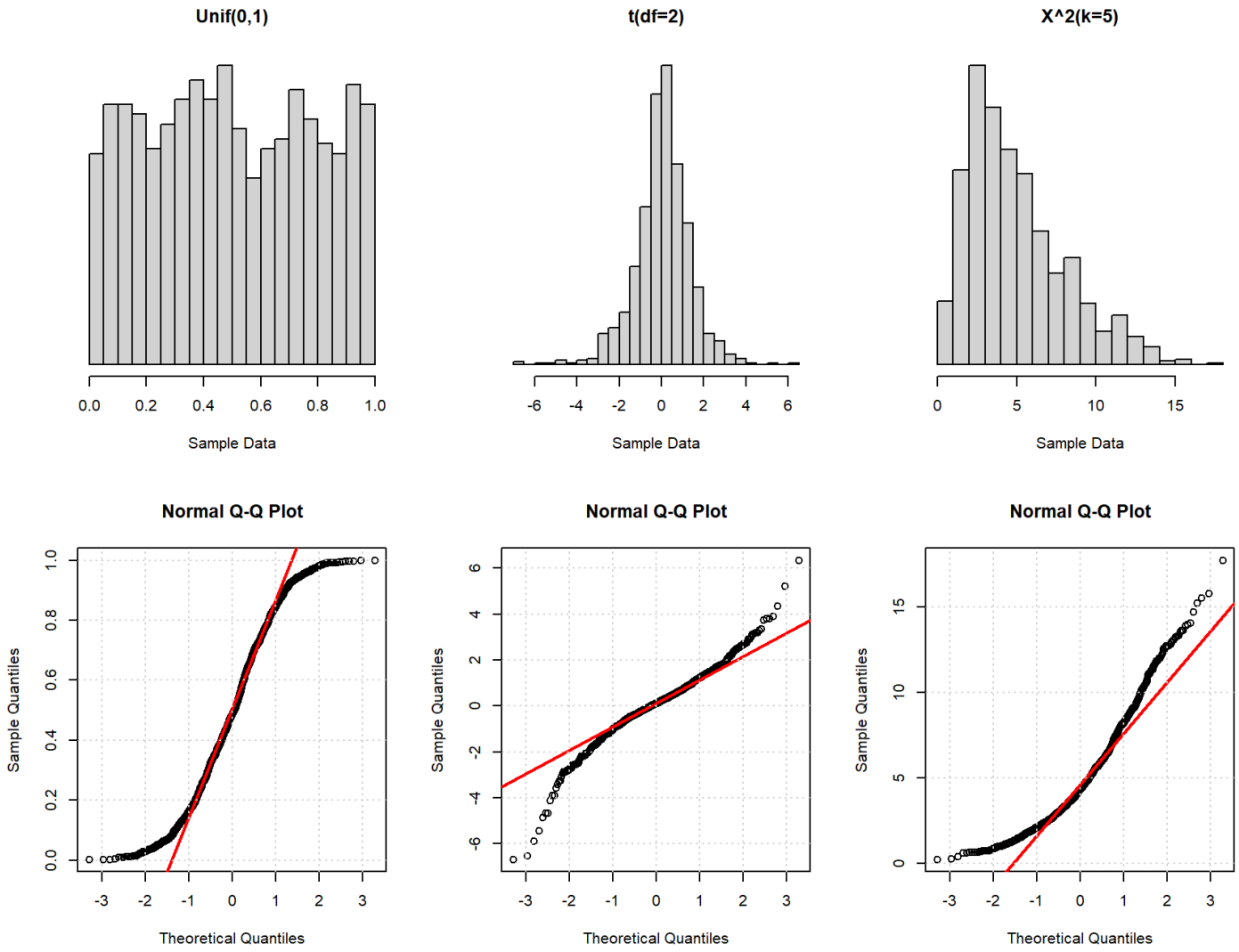
qqnorm(r)
qqline(r, col = 'red', lwd = 2)
grid()
```





- If the residuals are normally distributed, then the points on the Q-Q plot should fall along a straight line. (This line does not need to be exactly the diagonal $y = x$ line.)

- Note that a histogram of the observed responses y_i's does not necessarily have a bell-shape. Although each response random variable $Y_i \sim \mathcal{N}(\mu_i, \sigma^2)$ is normally distributed, each has its own expectation $\mathbb{E}(Y_i) = \mu_i$. So they are not a sample from one common normal distribution.
- Below are some examples of Q-Q plot which compare data samples from a thin-tail $\text{Unif}(0, 1)$ distribution, a fat-tail $t_{\nu=2}$ distribution, and a right-skewed $\mathcal{X}^2_{k=5}$ distribution, each against the standard normal $\mathcal{N}(0, 1)$ distribution. Note the specific ways these plots deviate from a straight line.



- There are also many normality tests (https://en.wikipedia.org/wiki/Normality_tests) developed by statisticians over the years. See R's `library(nortest)` . e.g.
  - Kolmogorov–Smirnov Test (https://en.wikipedia.org/wiki/Kolmogorov%E2%80%93Smirnov_test) (1939) `ks.test()`
  - Anderson–Darling test (https://en.wikipedia.org/wiki/Anderson%E2%80%93Darling_test) (1952)
  - Shapiro–Wilk test (https://en.wikipedia.org/wiki/Shapiro%E2%80%93Wilk_test) (1965) `shapiro.test()` and its variant Shapiro–Francia test (https://en.wikipedia.org/wiki/Shapiro%E2%80%93Francia_test) (1972)
  - Lilliefors test (https://en.wikipedia.org/wiki/Lilliefors_test) (1967)
  - Jarque–Bera test (https://en.wikipedia.org/wiki/Jarque%E2%80%93Bera_test) (1980)

```
x <- rnorm(100, mean = 2, sd = 3)

ks.test(x, pnorm, mean = mean(x), sd = sd(x)) # Need to specific the exact reference distribution
shapiro.test(x)
```

```
    Asymptotic one-sample Kolmogorov-Smirnov test

data:  x
D = 0.047014, p-value = 0.9799
alternative hypothesis: two-sided


    Shapiro-Wilk normality test

data:  x
W = 0.9956, p-value = 0.9876
```

```
x <- runif(100)

ks.test(x, pnorm, mean = mean(x), sd = sd(x))
shapiro.test(x)
```

```
    Asymptotic one-sample Kolmogorov-Smirnov test

data:  x
D = 0.089053, p-value = 0.4059
alternative hypothesis: two-sided


    Shapiro-Wilk normality test

data:  x
W = 0.95935, p-value = 0.003613
```

```
x <- rt(100, df = 2)

ks.test(x, pnorm, mean = mean(x), sd = sd(x))
shapiro.test(x)
```

```
    Asymptotic one-sample Kolmogorov-Smirnov test

data:  x
D = 0.32428, p-value = 1.47e-09
alternative hypothesis: two-sided


    Shapiro-Wilk normality test

data:  x
W = 0.4308, p-value < 2.2e-16
```

```
x <- rchisq(100, df = 5)
ks.test(x, pnorm, mean = mean(x), sd = sd(x))
shapiro.test(x)
```

```
    Asymptotic one-sample Kolmogorov-Smirnov test

data:  x
D = 0.081988, p-value = 0.5122
alternative hypothesis: two-sided


    Shapiro-Wilk normality test

data:  x
W = 0.97129, p-value = 0.02779
```

- What is the issue with violating the normality assumption?
    - If the violation is due to some large (positive or negative) residuals i.e. outliers, then we should revisit our fitted model. Perhaps we should consider other robust regression (https://en.wikipedia.org/wiki/Robust_regression) techniques that are less sensitive to outliers.
    - Improper calibration of confidence interval, hypothesis testing, and p-values.